

Part A - Theoretical Foundation

1) Inferential Statistics

Inferential statistics is the branch of statistics that uses sample data to make conclusions or predictions about a whole population.

Formula idea :

$$\text{Population mean } (\mu) \approx \text{Sample mean } (\bar{x})$$

2) Hypothesis Testing & Components

Hypothesis testing is a statistical method used to check whether a claim about a population is true using sample data.

Main components:

- Null hypothesis $\rightarrow H_0$
- Alternative hypothesis $\rightarrow H_1$
- Significance level $\rightarrow \alpha$
- Test statistic

Common test statistic formula (z-test):

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

3) Confidence Interval & Critical Value

A confidence interval gives a range of values that likely contains the true population parameter. A critical value is the cutoff point used to decide whether to reject the hypothesis.

Confidence interval formula:

$$\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

4) p-value

The p-value is the probability of getting a result as extreme as the observed data, assuming the null hypothesis is true.

Decision rule:

$$\text{Reject } H_0 \text{ if } p \leq \alpha$$

5) Type I and Type II Errors

Type I error happens when we reject a true null hypothesis.

Type II error happens when we fail to reject a false null hypothesis.

Formulas:

$$P(\text{Type I error}) = \alpha$$

$$P(\text{Type II error}) = \beta$$

$$\text{Power of test} = 1 - \beta$$

6) z-test, t-test, Chi-square test, ANOVA

z-test: Used when population variance is known and sample size is large.

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

t-test: Used when population variance is unknown and sample size is small.

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Chi-square test: Used to compare observed and expected categorical data.

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

ANOVA: Used to compare means of more than two groups.

$$F = \frac{\text{Between-group variance}}{\text{Within-group variance}}$$

7) Covariance

Covariance shows how two variables change together. Positive means they increase together, negative means one increases while the other decreases.

Formula:

$$\text{Cov}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n - 1}$$

8) Correlation

Correlation measures the strength and direction of the relationship between two variables. Its value lies between -1 and $+1$.

Formula:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y}$$